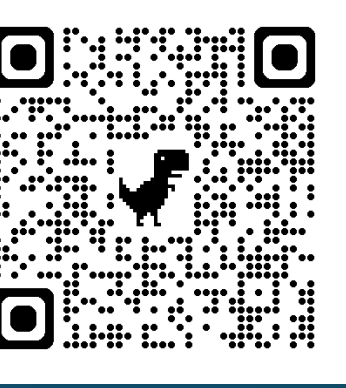


Robustness of Bayesian Exponentially Tilted Empirical Likelihood

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1. Introduction

- Moment condition models avoid the need to specify the full data distribution process.
- Combine advantages of EL + ET $\rightarrow$ ETEL, which can be interpreted probabilistically as a nonparametric likelihood.
- ETEL provides a useful framework for Bayesian moment condition models (Chib et al., 2018).

2. Methodology

Moment condition model:

$$E_P[g(\mathbf{y}, \boldsymbol{\theta})] = \mathbf{0} \quad (1)$$

- Parameter of interest $\boldsymbol{\theta} := (\theta_1, \dots, \theta_p)$.
- Observed data $\mathbf{y} = (y_1, \dots, y_n)$ from the unknown distribution P .

ETEL (Schennach, 2005):

- Defined as

$$p_{ETEL}(\mathbf{y}|\boldsymbol{\theta}) = \prod_{i=1}^n p_i(\boldsymbol{\theta}) \quad (2)$$

- By solving the constrained maximization problem:

$$\max_{p_1, \dots, p_n} \sum_{i=1}^n [-p_i \log(np_i)] \quad (3)$$

subject to

$$\sum_{i=1}^n p_i(\boldsymbol{\theta}) = 1 \text{ and } \sum_{i=1}^n p_i(\boldsymbol{\theta}) g(y_i; \boldsymbol{\theta}) = \mathbf{0} \quad (4)$$

- The ETEL probabilities:

$$p_i(\boldsymbol{\theta}) := \frac{\exp(\hat{\lambda}(\boldsymbol{\theta})^T g(y_i; \boldsymbol{\theta}))}{\sum_{j=1}^n \exp(\hat{\lambda}(\boldsymbol{\theta})^T g(y_j; \boldsymbol{\theta}))} \quad (5)$$

where the Lagrange multipliers $\hat{\lambda}(\boldsymbol{\theta})$ is given by

$$\hat{\lambda}(\boldsymbol{\theta}) = \arg\min_{\lambda} \sum_{i=1}^n \exp(\lambda^T g(y_i; \boldsymbol{\theta})) \quad (6)$$

Bayesian ETEL (BETEL) (Schennach, 2007):

- Through Bayes' theorem, the BETEL posterior takes the form:

$$\pi_{BETEL} \propto p_{ETEL}(\mathbf{y}|\boldsymbol{\theta})\pi(\boldsymbol{\theta}) \quad (7)$$

3. Motivating Example – How does BETEL perform in the presence of outliers?

(1) Setup – Simple linear regression

$$y_i = \beta_0 + \beta_1 x_i + e_i$$

- $\beta_0 = 2, \beta_1 = -3; x_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(1,1); n = 1,000; 50$ replications
- Moment conditions:

$$E_P[e_i] = 0, \quad E_P[x_i e_i] = 0, \quad E_P[e_i^3] = 0. \quad (8)$$

- Data contamination:

$$H_e = (1 - 0.10)\mathcal{N}(0,1) + 0.10 \mathcal{N}(5,1) \quad (9)$$

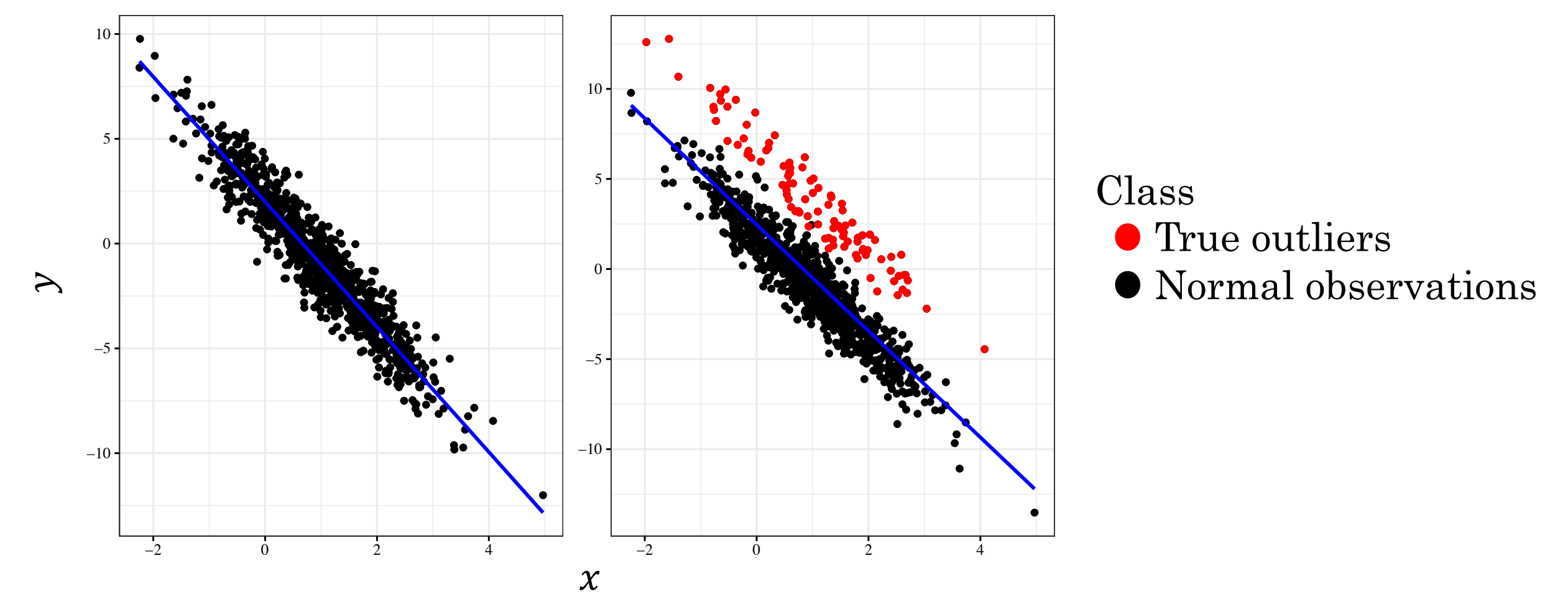


Figure 1: Simulated datasets.

(2) MCMC computation

- Metropolis-Hasting within Gibbs with adaptMCMC (R Package adaptMCMC, Scheidegger, 2024; Vihola, 2012)
- Acceptance rate: 0.234; 50,000 draws; 20,000 burn-in; β_0 & β_1 sampled jointly

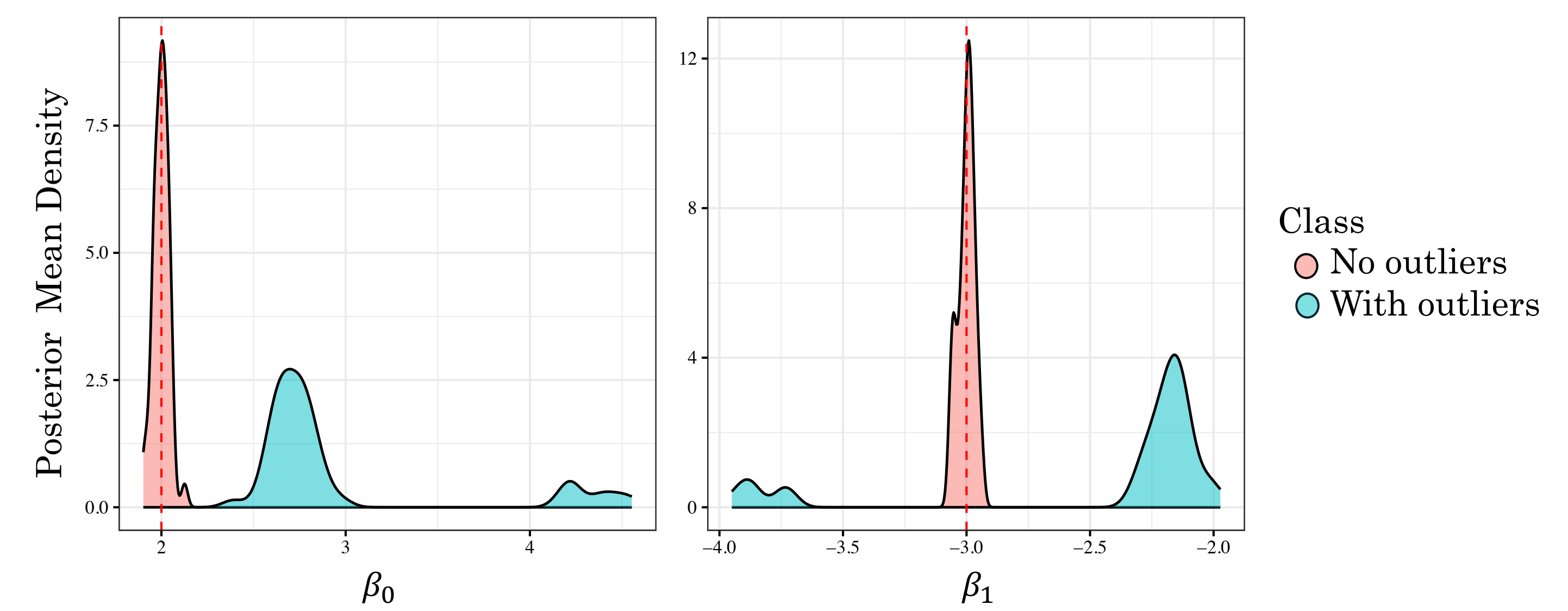


Figure 2: BETEL posterior means distributions across 50 replications.

4. Diagnostics for Outliers – BETEL is sensitive to outliers, can we turn this into a feature to detect them?

(1) Local sensitivity via KL divergence (Millar and Stewart, 2007)

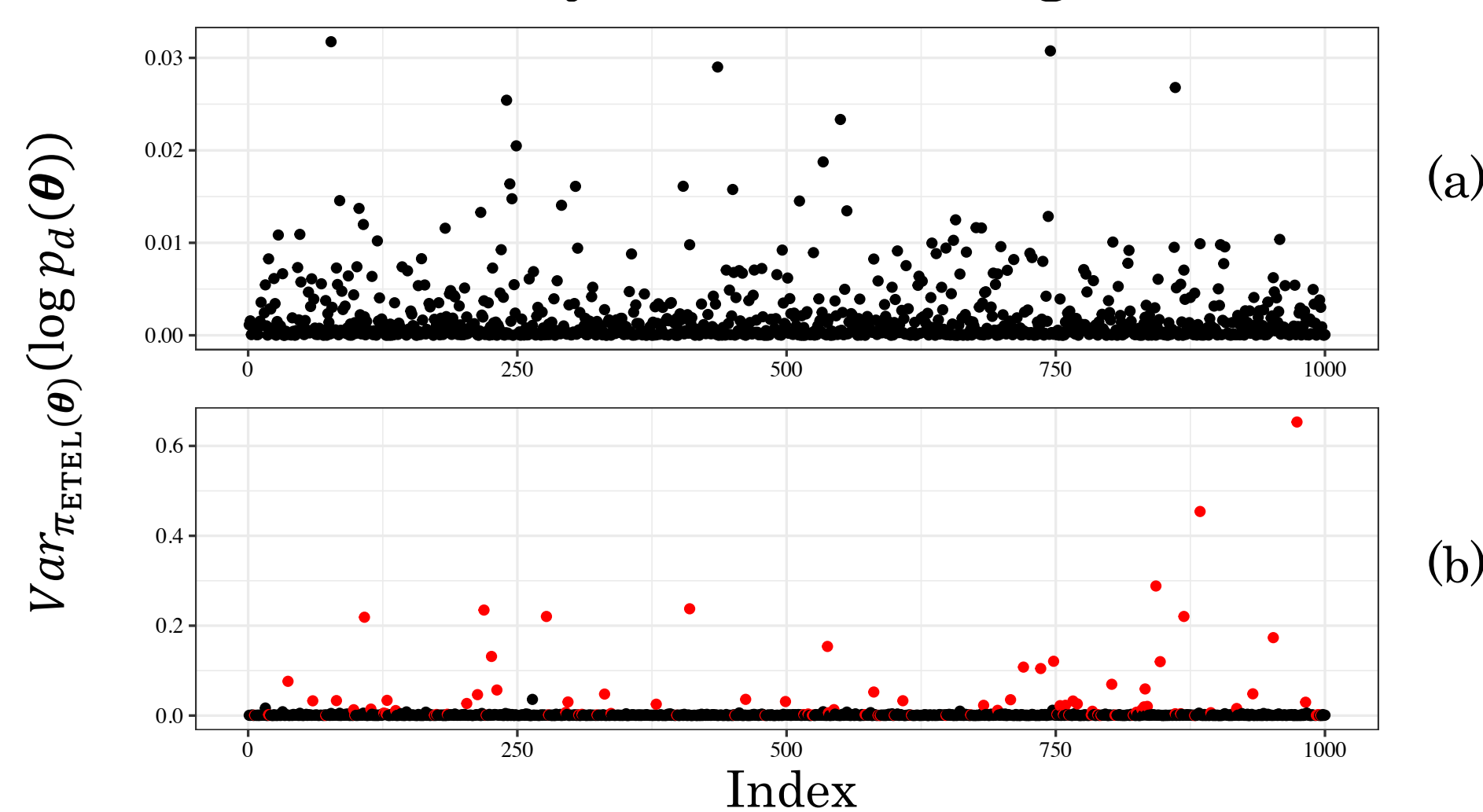


Figure 3: Local sensitivity (KL) (a) without and (b) with outliers.

(2) Standardized pseudo-residuals (Zhu et al., 2008)

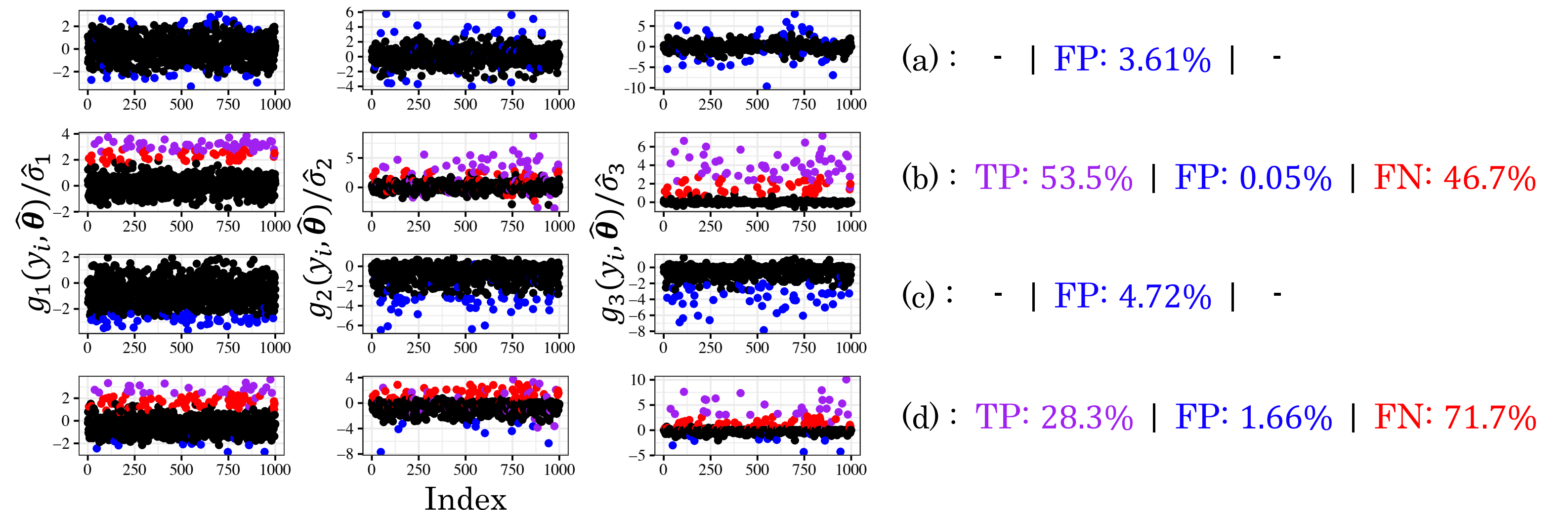
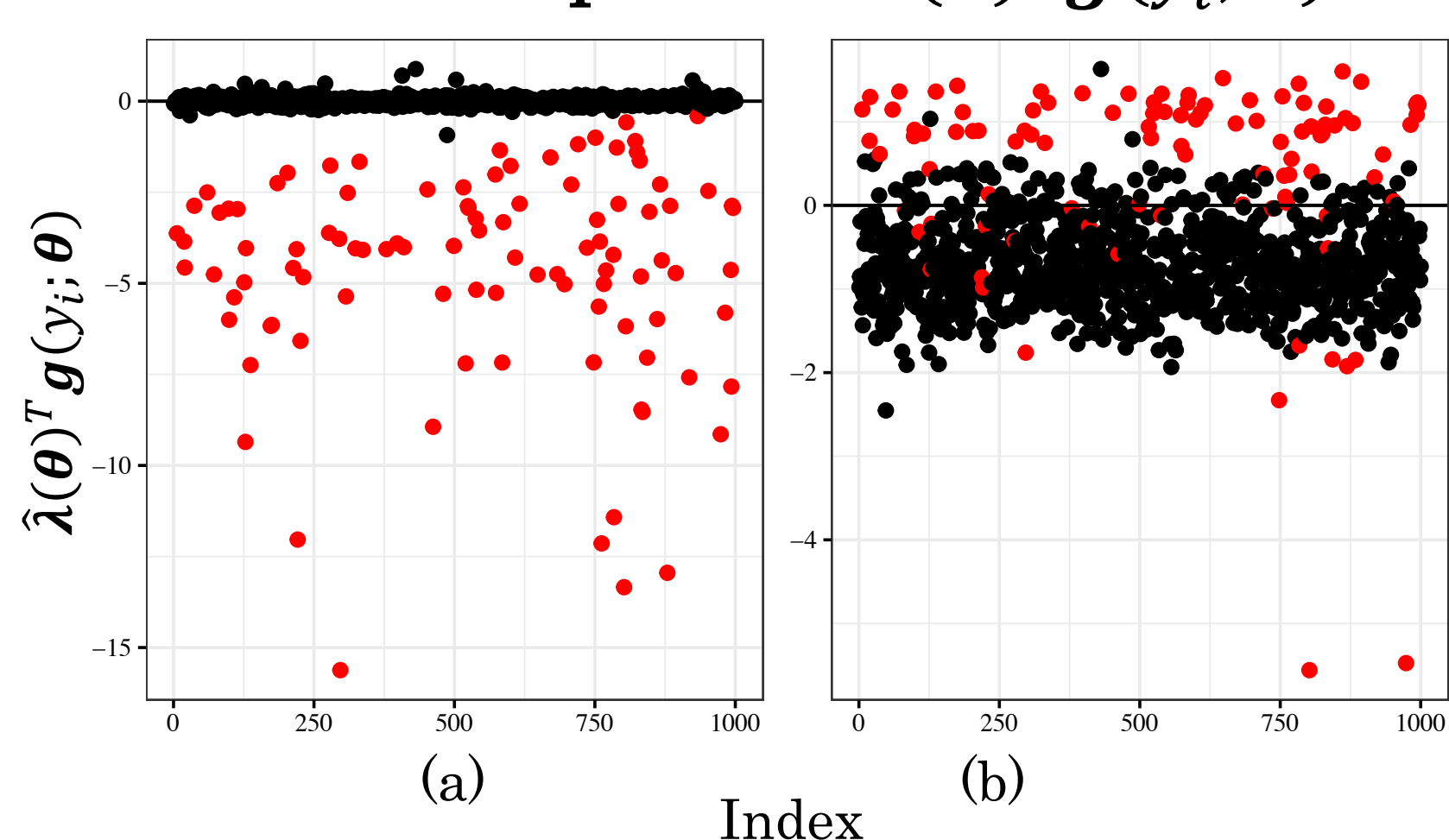


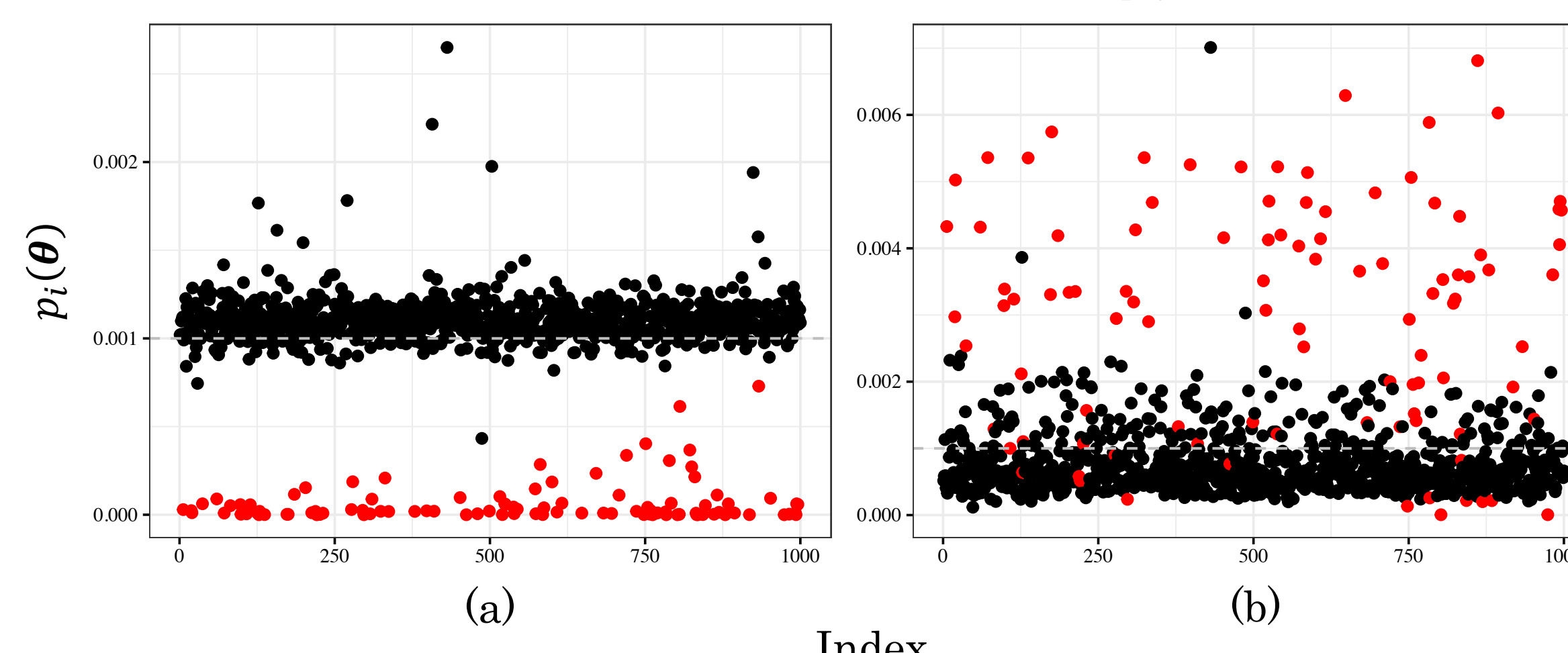
Figure 4: Residuals at different β values (left) and detection rates (right).

5. Mechanism of ETEL – Why does it fail to detect outliers?

(1) Inner product $\hat{\lambda}(\boldsymbol{\theta})^T g(y_i; \boldsymbol{\theta})$



(2) ETEL probabilities $p_i(\boldsymbol{\theta})$



(3) Log-likelihood $\sum \log(p_i(\boldsymbol{\theta}))$

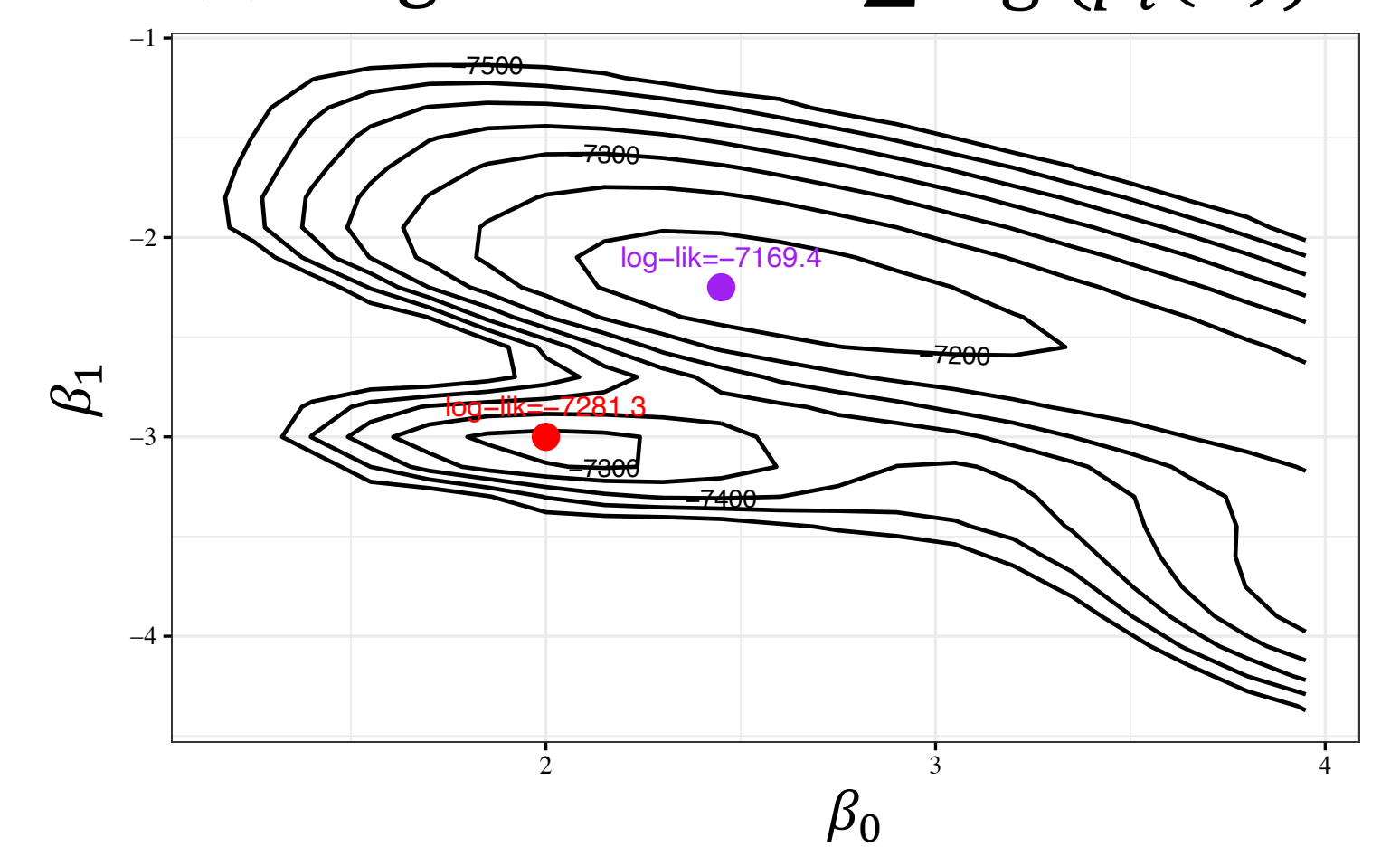


Figure 5: (1) Inner product and (2) ETEL probabilities at true vs. biased parameters. (3) Log-likelihood contour.

Black: normal obs; Red: true outliers in (1, 2) / log-likelihood at true parameters in (3); Purple: maximum log-likelihood; Likelihood gap (max – true) = +111.9.

6. Contributions and Future Research

Contributions:

- Illustrate BETEL's non-robustness: log-likelihood collapses under data contamination, biasing posteriors.
- Show that posteriors place no mass on the truth and appear normal, leaving practitioners unaware that inference is unreliable.
- Demonstrate diagnostics can flag outliers but only at or near the truth.

Future research:

- Develop tools to flag biased estimates.
- Test on real-world data.

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